

In the Clinic®

## Urinary Tract Infection

Urinary tract infections (UTIs) are common in both inpatient and outpatient settings. This article provides an evidence-based, clinically relevant overview of management of UTIs, including screening, diagnosis, treatment, and prevention. Conditions covered include acute cystitis (both uncomplicated and complicated), catheter-associated UTI, and asymptomatic bacteriuria in both women and men.

Screening and Prevention

Diagnosis and Evaluation

Treatment and Management

Practice Improvement

CME/MOC activity available at [Annals.org](https://annals.org).

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Urinary tract infections (UTIs) are common in both outpatient and inpatient settings. Clinical entities encompassed by the term “UTI” include asymptomatic bacteriuria (ASB), acute uncomplicated cystitis, recurrent cystitis, catheter-associated ASB, catheter-associated UTI (CAUTI), prostatitis, and pyelonephritis. These categories are further distinguished by the presence or absence of symptoms referable to the urinary tract and the patient's sex, comorbid conditions, and genitourinary history, including the presence of stones, stents, or catheters. Because acute cystitis is the most common manifestation of UTI and is most prevalent in women, most clinical research on UTI has been done in women. Clinicians must consider whether recommendations derived from this evidence base are applicable to their patient populations.

In the absence of known abnormalities of the urinary tract, women are at higher risk for UTIs than men. Premenopausal women are at especially high risk for acute cystitis—incidence is 0.5–0.7 case per person-year among sexually active women (1). Other populations at risk for UTI include patients with voiding abnormalities related to diabetes; neurogenic bladder; spinal cord injury; pregnancy; prostatic hypertrophy; or urinary tract instrumentation, including

long-term ( $\geq 30$  days) indwelling catheters. As men age, acquired abnormalities of the urinary tract impair normal bladder emptying, which narrows the sex-based difference in UTI rates (2).

The strongest risk factors in premenopausal women include sexual intercourse, use of spermicides, and previous UTI. A history of maternal UTI and age at first UTI are also important risk factors in this group, suggesting a genetic component to susceptibility (3). Pregnant women are also at increased risk. Changes in vaginal microbial flora in perimenopausal women may increase risk for UTI. In contrast to the predominant role of behavioral risk factors in premenopausal women, mechanical and physiologic factors that affect bladder emptying are important in postmenopausal women (4). Diabetes increases risk for all urinary infectious disorders, from ASB to perirenal abscess and emphysematous pyelonephritis (5). In men, risk for UTI is primarily related to prostatic hypertrophy that occurs with advancing age. Temporary urinary catheterization is the major medical intervention that increases risk for UTI. Other comorbid conditions that increase risk in both sexes include the presence of stones or foreign bodies (such as ureteral stents) in the urinary system and diseases associated with neurogenic bladder.

## Screening and Prevention

### Is there a role for screening for UTI or ASB?

In men and nonpregnant women, screening for ASB is generally not recommended because treatment does not improve clinical outcomes (6). ASB does not lead to hypertension, chronic kidney disease, or decreased duration of survival (7). Women with ASB are at increased risk for symptomatic UTI, but treatment

does not decrease this risk (8). ASB is a marker for poor overall health status in diabetic patients, noncatheterized women in retirement homes, and catheterized inpatients but is not an independent risk factor for death.

ASB during pregnancy (4%–7% of pregnant women) is associated with progression to symptomatic UTI, including pyelonephritis (9). Acute pyelonephritis occurs in up

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to 2% of pregnancies in the United States and is the most common nonobstetric cause of hospitalization during pregnancy (9). ASB during pregnancy is also associated with low birthweight and preterm labor, although a causative relationship has not been established.

*The most recent Cochrane review (10) on this topic found that antibiotic treatment of ASB was associated with significantly decreased risk for pyelonephritis, premature birth, and low birthweight compared with no treatment. However, the 14 included studies were done between 1960 and 1987, and the risk of bias was judged to be high. The quality of the evidence was judged to be very low (pyelonephritis) and low (prematurity and low birthweight).*

*A cohort study of 4283 healthy pregnant women in the Netherlands who had very low risk for preterm delivery found much lower rates of pyelonephritis than previous studies (11). The proportion of pregnant women with untreated ASB who developed pyelonephritis was 5 of 208 (2.4%) compared with 24 of 4034 (0.6%) without ASB (odds ratio [OR], 3.9 [95% CI, 1.4–11.4]). Women with untreated ASB were also more likely than those without ASB to have subsequent UTI treated with antibiotics during pregnancy (OR, 2.9 [CI, 2.0–4.2]) and recurrent UTI treated with antibiotics during pregnancy (OR, 3.5 [CI, 1.8–6.7]).*

Screening for and treatment of ASB are also recommended before transurethral resection of the prostate (TURP) or other urinary tract instrumentation resulting in mucosal bleeding. Studies have shown that TURP in bacteriuric men can precipitate bacteremia with associated sepsis and that antimicrobial treatment of the bacteriuria can prevent these complications. The level of risk associated with specific invasive urologic procedures other than TURP in patients with preexisting bacteriuria is not well-defined. However, the Infectious Diseases Society of America (IDSA) recommends that procedures anticipated to cause mucosal bleeding warrant screening by urine culture and treatment of ASB before the procedure (6). Simple catheter

placement and cystoscopy without biopsy do not warrant screening for ASB.

Renal transplant recipients who have ASB are at higher risk for pyelonephritis, but whether pyelonephritis affects graft function is controversial (12, 13). The relationship of ASB to UTI and sepsis in patients with neutropenia has not been well-studied. Bacteriuria diagnosed in neutropenic patients as part of a fever work-up is by definition symptomatic.

### How can UTI be prevented?

Prevention of symptomatic UTI in patients with ASB is recommended only in pregnant women or before an invasive urologic procedure. Urine culture should be done and infection should be treated with directed antimicrobial therapy based on culture results.

Symptomatic UTI in women with recurrent episodes can be prevented with antimicrobial prophylaxis. This decision should be individualized. Given increasing antimicrobial resistance, regular use of antimicrobials to prevent UTI may not be appealing, and approaches other than antimicrobial prophylaxis, such as patient-initiated therapy or nonantimicrobial prevention, may be preferred even if they are less effective in reducing symptomatic events. If antibiotics are used to prevent recurrent UTI, they may be prescribed postcoitally or continuously. Postcoital antibiotic prophylaxis has been shown to be highly effective in preventing symptomatic recurrences in women with 3–4 UTIs per year, particularly if these are temporally associated with coitus.

*A randomized, double-blind, placebo-controlled trial found that among women with at least 2 culture-documented UTIs in the previous year, postcoital use of a single dose of trimethoprim-sulfamethoxazole (TMP-SMX) (40 mg plus 200 mg) resulted in an infection rate of 0.3 per patient per year compared with 3.6 per pa-*

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tient per year in the control group (14). Adverse effects were infrequent and minor. In women who are intolerant of or resistant to TMP-SMX, an alternative agent (such as nitrofurantoin macrocrystals or a fluoroquinolone) may be as effective as TMP-SMX for postcoital prophylaxis, although clinical evidence specifically related to postcoital use is not available for these agents.

For women with more frequent recurrences, more frequent coitus, or recurrences temporally unrelated to coitus, continuous (daily, thrice-weekly, or weekly) prophylaxis may be preferable, with dosing frequency tailored to the individual patient's response. However, rates decrease only during the active prophylaxis period and in many cases return to baseline levels after antimicrobial therapy is discontinued. Because some women may not wish to take prophylactic antibiotics regularly, patient-initiated therapy may have greater appeal.

*In a meta-analysis of 19 trials of prophylactic antibiotics for acute cystitis, 17 of which tested daily antibiotic use, the range of having 1 clinical recurrence was 0–0.27 person-years in the antibiotic group and 1.12–3.6 person-years in the placebo group (15).*

Increased awareness of the role of the microbiome in overall health, concern for selection for antibiotic-resistant bacteria, and the difficulty of treating resistant urinary pathogens have led to exploration of nonantimicrobial approaches to UTI prevention. These include probiotics (typically lactobacilli), cranberry products, D-mannose, and vaccines. A Cochrane meta-analysis of trials of lactobacilli concluded that probiotics did not significantly reduce risk for UTI compared with placebo or antibiotics (16). Currently, no vaccines are approved for use in UTI prevention in the United States, although preclinical testing has identified promising candidates (17).

Debate is ongoing about the effectiveness of cranberry products in the prevention of UTI.

*A randomized trial found a potentially protective effect on recurrent UTI with cranberry products compared with placebo; however, the study was underpowered (18). Another randomized controlled trial of twice-daily cranberry juice consumption versus placebo found no between-group difference in the rate of UTI recurrence (19). In a recent randomized double-blind trial of women in nursing homes, no significant difference was observed for any of the clinical or microbiologic outcomes (20). In another trial that included premenopausal women with complicated and uncomplicated recurrent UTI, TMP-SMX was more effective at preventing recurrent UTI than cranberry capsules, but at the expense of promoting antibiotic resistance (21). Daily consumption of a cranberry beverage decreased the number of clinical UTIs in a randomized, double-blind, placebo-controlled, multicenter study that included women with a recent UTI (22). The number of antibiotic courses for symptomatic UTIs also was reduced in the cranberry group compared with the placebo group.*

*A Cochrane review and meta-analysis concluded that cranberry juice cannot be recommended for prevention of UTI (23). However, in another meta-analysis, cranberry-containing products were associated with a protective effect against UTIs (24). A protective effect of cranberry products was also found in a recent meta-analysis, although the patient populations in the studies that provided the summarized data were heterogeneous, comprising children, men with prostate cancer, and women with recurrent UTI. The cranberry products studied were also heterogeneous (25).*

In postmenopausal women, daily topical application of intravaginal estriol cream may help reduce the frequency of symptomatic episodes.

*Two studies in a 2008 Cochrane systematic review (26) compared vaginal estrogens with placebo and concluded that vaginal estrogens reduced the number of UTIs in postmenopausal women. Oral estrogens did not reduce UTI compared with placebo. Use of vaginal estrogens can result in vaginal itching, burning, discharge, and metrorrhagia (26). Clinical practice guidelines developed by the Society of Gynecologic Surgeons Systematic Review Group (27) recommend vaginal estrogens for postmenopausal women with recurrent UTI. Data on the risk for cancer in women using vaginal estrogens are limited (27). A discussion of possible cancer risk should be individualized for each patient.*

**Screening and Prevention...** Inappropriate screening for ASB can lead to unnecessary antibiotic use. Screening for and treatment of ASB are recommended only in pregnant women and patients about to have an invasive urologic procedure. Postcoital antibiotic prophylaxis should be considered for women with 2 or more UTIs per year, particularly if these are temporally associated with coitus. Otherwise, daily or thrice-weekly antibiotic prophylaxis can be used to prevent UTI. Clinicians should consider topical intravaginal estrogen therapy to prevent symptomatic UTIs for postmenopausal women with recurrent UTIs. Evidence on the usefulness of cranberry products for UTI prevention is inconclusive.

## CLINICAL BOTTOM LINE

## Diagnosis and Evaluation

### What signs and symptoms should raise suspicion of UTI?

The most common symptoms of UTI in noncatheterized persons include dysuria, urinary frequency, and urgency.

*A meta-analysis of the accuracy of history and physical examination for diagnosing acute uncomplicated UTI in ambulatory women found that the probability of UTI was approximately 50% in those who presented with 1 or more symptoms. Specific combinations of symptoms increased the probability to more than 90%. Symptoms that increased the probability were dysuria, hematuria, and costovertebral angle tenderness. A history of vaginal discharge or irritation decreased the probability of UTI (28).*

*A prospective trial enrolled women who had at least 2 episodes of cystitis in the prior 12 months from a university-based primary care clinic (29). The participants were given urine collection materials and a course of fluoroquinolone antibiotics, both to be used if they developed symptoms of UTI. Overall, 88 of 172 women self-diagnosed a total of 172 UTIs. Laboratory evaluation showed a uropathogen in 144 cases (84%), sterile pyuria in 19 cases (11%), and no pyuria or bacteriuria in 9 cases (5%). Clinical and microbiological cures occurred in 92% and 96%, respectively, of culture-confirmed episodes.*

In other populations, such as women visiting emergency departments, self-diagnosis of UTI has been found to be less accurate. Low agreement between patients and clinicians and a higher rate (17%–21%) of chlamydia diagnoses have been reported in studies evaluating ac-

curacy of cystitis self-diagnosis in emergency department settings (30, 31). Many women in these studies had multiple sex partners or vaginal discharge and thus did not meet criteria for patient-initiated therapy. Thus, this approach should be used only in settings with an established patient-provider relationship and low risk for sexually transmitted disease (STD).

In catheterized patients, signs and symptoms suggestive of CAUTI include new onset or worsening of fever, rigors, altered mental status, malaise or lethargy with no other identified cause, flank pain, costovertebral angle tenderness, acute hematuria, or pelvic discomfort. In patients whose catheters have been removed within the past 48 hours, dysuria, urgent or frequent urination, or suprapubic pain or tenderness are also suggestive of CAUTI (32). Guidelines also specify that  $10^3$  or more colony-forming units per milliliter of urine is sufficient for diagnosis of CAUTI. By definition, CAUTI can occur with indwelling urethral (Foley) catheters, suprapubic catheters, or condom catheters. Because many of these defining symptoms are nonspecific, other infections and potential causes should be considered before attributing the symptoms

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**Table 1. Differential Diagnosis of Acute Cystitis in Women**

| Condition      | Pathogens                                                                                        | History                                                                                                                                    | Symptoms                                                                                                                                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Vaginitis      | Candida, <i>Trichomonas vaginalis</i> , <i>Bacteroides</i> species, <i>Gardnerella vaginalis</i> | Possibly new sex partner or unprotected sexual activity; history of vaginitis                                                              | Vaginal discharge, odor, or itching; "external" dysuria (from urine coming into contact with inflamed and irritated vulvar epithelial surfaces)                                                                           |
| Urethritis     | <i>Chlamydia trachomatis</i> , <i>Neisseria gonorrhoeae</i> , or herpes simplex virus            | New sex partner, unprotected sexual activity, history of sexually transmitted disease or recurrent genital herpes simplex virus            | Gradual onset of symptoms ( <i>Chlamydia</i> ) ± vaginal discharge; ± urinary frequency or urgency                                                                                                                        |
| Irritation     | None                                                                                             | No unusual sexual exposure; possible chemical or allergen exposures (e.g., douches, bath products, feminine hygiene products, spermicides) | Vaginal itching or discharge; usually a diagnosis of exclusion, unless withdrawal of a suspected offending substance leads to resolution of symptoms                                                                      |
| Pyelonephritis | Same as acute cystitis                                                                           | Previous urinary tract infection (pyelonephritis or cystitis)                                                                              | Constitutional symptoms (fever, malaise, sweats, headache), gastrointestinal symptoms (anorexia, nausea, vomiting, abdominal pain), local renal symptoms (back, flank, or loin pain), ± voiding symptoms (as in cystitis) |

to catheter-associated bacteriuria. Distinguishing between CAUTI and catheter-associated ASB is challenging.

### What other disorders should be considered?

Alternative diagnoses to consider include sexually transmitted urethritis or vaginitis, noninfectious urethritis, and early pyelonephritis (**Table 1**). Symptoms of STDs may be subtle and thus should always be considered in the differential diagnosis. A history of vaginal discharge or irritation decreases the probability of UTI and warrants a work-up for STDs and other vaginal conditions, such as candidiasis (28).

*A cross-sectional study screened 296 sexually active females aged 14–22 years who were visiting a teen health center for both UTI (by urine culture) and STD (by vaginal swab and nucleic acid amplification testing). In this population, the prevalence of UTI and STD were 17% and 33%, respectively; 4% had both. The presence or absence of urinary symptoms did not predict STD, indicating that telephone management may not be appropriate for adolescent females with urinary symptoms (33).*

Clinicians should also consider pyelonephritis (or prostatitis in men) before initiating therapy for acute cystitis. The recommended duration of therapy for pyelone-

phritis is longer than for cystitis; treatment of incipient pyelonephritis with a short-course regimen for cystitis could predispose the patient to relapse. Pyelonephritis may or may not be associated with symptoms of cystitis, which in some patients may overshadow the renal or systemic manifestations. In patients presenting with symptoms of acute cystitis, clinicians should confirm the absence of fever, nausea, vomiting, rigors, and flank pain. In men with urinary symptoms and fever, both acute infectious prostatitis and pyelonephritis should be considered in the differential diagnosis.

### What tests should be done?

Women presenting with classic symptoms of cystitis (dysuria or urinary frequency) without symptoms suggesting alternative diagnoses or underlying complications may be treated for UTI without further testing. For such women, self-initiated therapy or "telephone clinic" provider-guided presumptive therapy may be appropriate (34). In women presenting with symptoms of acute cystitis, a positive urine dipstick result can help to confirm the diagnosis, but a negative

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result cannot rule out the diagnosis in women with a high pretest probability (28, 35). Thus, a dipstick test is not necessary if the history is clearly diagnostic of UTI (particularly in women with symptoms akin to previous episodes).

In women with acute dysuria, pyuria is a highly sensitive indicator of bacteriuria and can be used as a surrogate for urine culture (35). If the diagnosis is not clear-cut or additional complicating factors or alternative diagnoses are possible, a more thorough and broad-based diagnostic evaluation should be done.

Blood tests (including cultures) are not indicated in women with suspected cystitis unless they are needed to screen for alternative diagnoses or to assess the status of a known underlying medical condition (such as diabetes mellitus). Up to 30% of women with acute pyelonephritis may have secondary bacteremia, and identification of the organism by blood culture can be helpful in cases where antibiotics are started in advance of the urine culture. Diabetic women and renal transplant recipients have a higher incidence of secondary bacteremia with UTI, so blood cultures should be done if these patients have systemic signs of infection (36, 37).

Pretreatment urine culture is also considered standard of care if the patient is pregnant or male. In women, urine culture is generally not required in suspected acute uncomplicated cystitis because the spectrum of causative organisms is predictable and urine culture results often are not available until after completion of short-course empirical treatments.

The role of urine culture is to provide retrospective confirmation of the presence of bacteriuria, which in the correct clinical setting con-

firms the diagnosis of UTI, and to provide specific information about the organism and its antimicrobial susceptibility. Performing a culture (with susceptibility testing) on a pretreatment urine sample is appropriate for women with suspected cystitis if the diagnosis is not clear, if an unusual or antimicrobial-resistant organism is suspected, if the episode represents a suspected relapse or treatment failure, or if the patient's therapeutic options are limited by medication intolerance.

*Escherichia coli* is isolated in more than 90% of patients with uncomplicated cystitis and pyelonephritis. Other coliforms, including *Klebsiella* and *Proteus*, are less common. *Staphylococcus saprophyticus* causes uncomplicated cystitis and pyelonephritis in a small proportion (5%–10%) of otherwise healthy women. *E coli* is a predominant pathogen in complicated UTI, but other coliforms and enterococci are also common. CAUTI in patients with short-term catheters can be caused by *E coli* as well as by a spectrum of typical hospital-acquired pathogens, including *Klebsiella*, *Citrobacter*, *Enterobacter*, *Pseudomonas*, coagulase-negative staphylococci, enterococci, and *Candida*. Patients with long-term catheters typically have polymicrobial infections; in addition to the aforementioned pathogens, *Proteus*, *Morganella*, and *Providencia* are common (38).

Interpretation of urine culture results depends on the clinical context and the urinalysis findings; the threshold of bacteriuria required for the diagnosis of acute cystitis is not absolute. Even a low concentration ( $10^2$  colony-forming units per milliliter) of coliform bacteria in a urine sample from a woman with acute dysuria and pyuria often represents true (that is, from within the urinary tract) bacteriuria. Such

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organisms should not be dismissed as “contaminants” in a clinical setting suggestive of cystitis. On the other hand, even high concentrations ( $>10^5$  colony-forming units per milliliter) of nonpathogens may not reflect true bacteriuria if the urine specimen was not collected properly or was allowed to stand at room temperature before processing. Organisms other than coliform bacilli, *S saprophyticus*, and *Enterococcus* (for example, lactobacilli,  $\alpha$ -streptococci, and coagulase-negative staphylococci other than *S saprophyticus*) are usually considered contaminants in urine cultures from women with uncomplicated cystitis, whereas in complicated UTI almost any organism can be causative and must be seriously considered if the patient is symptomatic.

Limitations of urine sampling are now better recognized. In a study comparing paired specimens of midstream urine and urine collected by means of a urethral catheter obtained from women with acute cystitis, the presence of *E coli* in midstream urine was highly predictive of bladder bacteriuria, even at very low counts (39). In contrast, both enterococci and group B streptococci were isolated frequently from midstream urine but rarely from paired specimens of urine sampled from the bladder through catheterization (39). Misinterpretation of such cultures may result in undertreatment of low-quantity *E coli* infections or inappropriate treatment of infection with enterococci or group B streptococci identified in midstream urine. Thus, in women with symptoms of acute cystitis, a low concentration of *E coli* (alone or in conjunction with other bacteria) in a midstream urine specimen should not be considered a contaminant. In another study, a cul-

ture protocol able to isolate fastidious organisms provided a more complete description of clinically relevant uropathogens (40). Compared with enhanced quantitative urine culture, standard urine culture missed 67% of uropathogens overall and missed uropathogens in 50% of participants with severe urinary symptoms (40).

### Is there a role for diagnostic imaging?

For uncomplicated bladder infections, imaging studies (abdominal radiography, ultrasonography, computed tomography, and excretory urography) add little or no benefit but increase cost and cause delays and sometimes discomfort or complications for the patient. Such studies should be done only if pretest suspicion is high for an alternative diagnosis or an anatomical problem (such as a bladder obstruction or stone) that requires intervention. A recent study presented a clinical prediction rule consisting of a history of urolithiasis, a urine pH of 7.0 or greater, and/or renal insufficiency (estimated glomerular filtration rate  $\leq 40$  mL/min/ $1.73\text{ m}^2$ ) to be used at emergency departments for selective use of radiologic imaging in adults presenting with febrile UTI (41). This rule may lead to a 40% reduction in the number of imaging studies without missing clinically relevant outcomes, such as infection of the kidneys' collecting system and renal abscess (41). Although men with acute cystitis should be considered for further evaluation for urologic abnormalities, imaging studies for acute cystitis in men younger than 45 years or in older men without symptoms of voiding difficulties or hematuria may not be useful. The evidence base for male cystitis is limited (42).

38. Nicolle LE. Catheter-related urinary tract infection. *Drugs Aging*. 2005;22:627-39. [PMID: 16060714]
39. Hooton TM, Roberts PL, Cox ME, Stapleton AE. Voided midstream urine culture and acute cystitis in premenopausal women. *N Engl J Med*. 2013;369:1883-91. [PMID: 24224622]
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**Diagnosis and Evaluation...** In ambulatory adult women, patient history has a high predictive value for presence or absence of cystitis. Clinicians should consider a diagnosis of pyelonephritis (or prostatitis in men) before starting therapy for acute cystitis. Complicating factors that would affect the appropriate diagnostic work-up and course of therapy should also be considered. Urinalysis should be used to confirm the diagnosis in women with suspected UTI when the history alone is not diagnostic. Clinicians should always perform urine culture in patients with pyelonephritis, complicated UTI, or a history of failure of initial therapy or multidrug resistance. Empirical therapy should be initiated and then adjusted on the basis of the results of the urine culture (if it is done).

## CLINICAL BOTTOM LINE

## Treatment and Management

### What are the preferred treatments of UTI?

Treatment of UTI depends on host factors (such as sex, compromised immune system, or urologic abnormalities), severity of illness, and risk for multidrug resistance. The prevalence of resistance in community populations has increased and must be considered, even in outpatients.

In women with symptoms of cystitis without fever or signs of systemic infection, outpatient oral antibiotic therapy is recommended. Clinicians should ask about factors that may influence the choice of antimicrobial agent, including pregnancy and breastfeeding, use of other medications, drug allergy history, recent antibiotic therapy, other recent infections or positive culture results, and recent travel. Previous microbiology should also be reviewed because it can be useful in predicting multidrug resistance of the current pathogen (43). Also, complicating factors should be considered because management of uncomplicated cystitis differs from that of pyelonephritis and complicated UTI. The IDSA has published guidelines for treatment of uncomplicated cystitis and pyelonephritis, specifically in premenopausal, nonpregnant women with no

known urologic abnormalities or comorbid conditions (44). Recommendations for treatment that take into account current resistance rates and other populations have been published more recently (45, 46).

Management of UTI is affected by the increasing prevalence of resistant organisms and the potential for propagation of resistance among normal host flora with the use of broad-spectrum antibiotics. Four agents are recommended for first-line therapy: nitrofurantoin, TMP-SMX, pivmecillinam, and fosfomycin trometamol (44). Alternative agents include  $\beta$ -lactams and fluoroquinolones. Two important modifications to previous recommendations should be noted. First, TMP-SMX should be used with caution as a first-line empirical agent because resistance is widespread. Clinicians should consider use in the setting of known susceptibility based on previous microbiology or practice in a low-resistance prevalence area. Second, fluoroquinolones have been moved to the last class of agents (after  $\beta$ -lactams) and should be used only when no other oral options are available because of a warning from the U.S. Food and Drug Administration (FDA) that risk for serious harms outweighs

43. Linsenmeyer K, Strymish J, Gupta K. Two simple rules for improving the accuracy of empiric treatment of multidrug-resistant urinary tract infections. *Antimicrob Agents Chemother*. 2015;59:7593-6. [PMID: 26416859]
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45. Walker E, Lyman A, Gupta K, Mahoney MV, Snyder GM, Hirsch EB. Clinical management of an increasing threat: outpatient urinary tract infections due to multidrug-resistant uropathogens. *Clin Infect Dis*. 2016;63:960-5. [PMID: 27313263]
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Table 2. Treatment Regimens for Acute Uncomplicated Cystitis

| Drug                                     | Dose and Duration                            | Common Adverse Effects                                                                 | Comments*                                                                                                                                                                                                                                                                            |
|------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Recommended agents</b>                |                                              |                                                                                        |                                                                                                                                                                                                                                                                                      |
| Nitrofurantoin monohydrate/macrocrystals | 100 mg twice daily for 5 d                   | Nausea, headache                                                                       | Resistance rare to date; may be useful for multidrug-resistant pathogens; cost varies; usually well-tolerated; FDA pregnancy category B                                                                                                                                              |
| Trimethoprim-sulfamethoxazole            | 160/800 mg (1 DS tablet) twice daily for 3 d | Rash, urticaria, nausea, vomiting, hematologic signs                                   | Excellent efficacy if local resistance <20%; resistance prevalence is increasing; use with caution unless known susceptibility; inexpensive; extensive clinical experience; avoid during pregnancy, particularly first and third trimesters; FDA pregnancy category C                |
| Fosfomycin trometamol                    | 3-g single-dose sachet                       | Diarrhea, nausea, headache                                                             | May be useful for multidrug-resistant pathogens; may be less effective than other agents; FDA pregnancy category B                                                                                                                                                                   |
| <b>Alternative agents</b>                |                                              |                                                                                        |                                                                                                                                                                                                                                                                                      |
| $\beta$ -Lactams                         | Dose varies by agent; 5- to 7-d regimen      | Diarrhea, nausea, vomiting, rash, urticaria                                            | Resistance varies by agent; increased adverse effects compared with other choices; FDA pregnancy category B                                                                                                                                                                          |
| Fluoroquinolones                         | Dose varies by agent; 3-d regimen            | Nausea, vomiting, diarrhea, headache, drowsiness, insomnia, tendon rupture, neuropathy | Risk may outweigh benefit for treatment of outpatient uncomplicated cystitis (FDA warning). Resistance prevalence increasing; cost varies; excellent efficacy; high collateral damage; better reserved for more serious conditions; avoid during pregnancy; FDA pregnancy category C |

DS = double-strength; FDA = U.S. Food and Drug Administration.

\* FDA pregnancy category B: Animal reproduction studies have failed to demonstrate a risk to the fetus and there are no adequate and well-controlled studies in pregnant women. FDA pregnancy category C: Animal reproduction studies have shown an adverse effect on the fetus and there are no adequate and well-controlled studies in humans, but potential benefits may warrant use of the drug in pregnant women despite potential risks ([www.drugs.com/pregnancy-categories.html](http://www.drugs.com/pregnancy-categories.html)).

benefits. Treatment regimens for cystitis and pyelonephritis are listed in **Tables 2** and **3**, respectively.

Each agent comes with caveats. Nitrofurantoin concentrates in urine but has little tissue penetration and should be avoided if

Table 3. Oral Treatment Regimens for Acute Uncomplicated Pyelonephritis\*

| Drug                                              | Dose and Duration                                                                           | Comments                                                                                                       | Other                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Fluoroquinolones</b>                           |                                                                                             |                                                                                                                |                                                                                                                                                                                                                                                                                                                                       |
| Ciprofloxacin<br>Ciprofloxacin XR<br>Levofloxacin | 500 mg twice daily for 5-7 d<br>1000 mg once daily for 5-7 d<br>750 mg once daily for 5-7 d | If local resistance prevalence is <10%                                                                         | Consider adding an initial 1-time intravenous dose of a long-acting parenteral antimicrobial (such as 1 g of ceftriaxone, 1 g of ertapenem, or a single 24-h dose of an aminoglycoside) if there is concern that the organism might be resistant, if the patient might not tolerate oral therapy, or if oral therapy could be delayed |
| Trimethoprim-sulfamethoxazole                     | 160/800 mg (1 DS tablet) twice daily for 10-14 d                                            | If pathogen known to be susceptible; otherwise, give an initial intravenous agent                              | Same as above                                                                                                                                                                                                                                                                                                                         |
| $\beta$ -Lactams                                  | Dose varies by agent; 10- to 14-d regimen                                                   | Oral $\beta$ -lactams are less effective and should be used with caution only when other agents cannot be used | Give an initial intravenous dose of a long-acting parenteral antimicrobial when using oral $\beta$ -lactams                                                                                                                                                                                                                           |

DS = double-strength; XR = extended-release.

\* For pyelonephritis, a urine culture and susceptibility testing should always be performed, and the initial empirical therapy should be tailored appropriately on the basis of the infecting uropathogen. Uncomplicated pyelonephritis indicates absence of pregnancy, nephrolithiasis, or urinary tract obstruction. The choice between these agents should be based on local resistance data, and the regimen should be tailored on the basis of susceptibility results.

there is a possibility of pyelonephritis. A 5-day course of nitrofurantoin was as effective as a 3-day course of TMP-SMX in terms of both clinical and microbiological cure in a randomized trial (47). The main limitation of use of TMP-SMX is the increasing rate of resistance among uropathogens and evidence that in vitro resistance correlates with bacterial and clinical failure in at least 50% of women. Fosfomycin is available in the United States but is used infrequently. The recommended 3-g single dose may not be as efficacious as other recommended agents, and it should not be used if pyelonephritis is suspected. Susceptibility data are not routinely reported for this drug, but surveys show that it retains activity against multidrug-resistant uropathogens, such as extended-spectrum  $\beta$ -lactamase-producing gram-negative organisms (48–50). Pivmecillinam, which is currently unavailable in the United States, is an extended-spectrum penicillin that is active against gram-negative organisms but is used only for treatment of UTI. Although efficacy rates are lower than for other recommended agents, its low resistance rates have made it a popular first-line choice in some European countries.  $\beta$ -Lactams generally have inferior efficacy and greater rates of adverse effects and thus are considered alternative agents. The fluoroquinolones ofloxacin, ciprofloxacin, and levofloxacin are highly efficacious in 3-day regimens but are recommended as alternative agents because of the aforementioned FDA warning. Increasing rates of fluoroquinolone resistance and concerns about safety reinforce the importance of reserving these agents for conditions other than acute cystitis.

Most data on treatment duration are from women with uncomplicated

cystitis. Whether patients with diabetes need longer treatment has not been robustly studied. In a retrospective observational cohort study, duration of therapy for women with diabetes and UTI was longer, but without improved outcomes compared with women without diabetes, suggesting that the duration of UTI therapy in women with diabetes should be the same as in those without (51). In men with UTI, data on the optimal duration are limited; however, an observational study of 14 versus 7 days found no benefit and higher rates of *Clostridium difficile* infection in the longer-treatment group (52).

Pyelonephritis is a tissue-invasive disease, and the initial empirical regimen should be broad enough to ensure in vitro activity against the likely uropathogen. Performing urine culture and susceptibility testing before starting therapy is essential so that therapy can be tailored. Clinicians should rule out complicated infection (pregnancy, nephrolithiasis, or obstruction), determine whether the patient can take oral antibiotics and whether they require hospitalization, and decide whether there is high risk for fluoroquinolone or extended-spectrum  $\beta$ -lactamase resistance (53).

If oral therapy is feasible, a 7-day regimen of oral ciprofloxacin is preferred if local resistance rates to fluoroquinolones do not exceed 10% (a threshold that is increasingly being exceeded, even in community-based pyelonephritis [53]). Giving the extended-release formulation of ciprofloxacin or a once-daily dose of levofloxacin for 5–7 days is acceptable, but evidence is limited (44). TMP-SMX is also effective if the pathogen is susceptible, but a course of 10–14 days is recommended. If susceptibility of the uropathogen is not known, an initial 1-time intrave-

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54. Chenoweth CE, Gould CV, Saint S. Diagnosis, management, and prevention of catheter-associated urinary tract infections. *Infect Dis Clin North Am*. 2014;28:105-19. [PMID: 24484578]
55. Talan DA, Stamm WE, Hooton TM, Moran GJ, Burke T, Irvani A, et al. Comparison of ciprofloxacin (7 days) and trimethoprim-sulfamethoxazole (14 days) for acute uncomplicated pyelonephritis in women: a randomized trial. *JAMA*. 2000;283:1583-90. [PMID: 10735395]
56. Hicks LA, Bartoces MG, Roberts RM, Suda KJ, Hunkler RJ, Taylor TH Jr, et al. US outpatient antibiotic prescribing variation according to geography, patient population, and provider specialty in 2011. *Clin Infect Dis*. 2015;60:1308-16. [PMID: 25747410]
57. Fleming-Dutra KE, Hersh AL, Shapiro DJ, Bartoces M, Enns EA, File TM Jr, et al. Prevalence of inappropriate antibiotic prescriptions among US ambulatory care visits, 2010-2011. *JAMA*. 2016;315:1864-73. [PMID: 27139059]
58. Grigoryan L, Zoorob R, Wang H, Trautner BW. Low concordance with guidelines for treatment of acute cystitis in primary care. *Open Forum Infect Dis*. 2015;2:ofv159. [PMID: 26753168]
59. May L, Mullins P, Pines J. Demographic and treatment patterns for infections in ambulatory settings in the United States, 2006-2010. *Acad Emerg Med*. 2014;21:17-24. [PMID: 24552520]
60. Shapiro DJ, Hicks LA, Pavia AT, Hersh AL. Antibiotic prescribing for adults in ambulatory care in the USA, 2007-09. *J Antimicrob Chemother*. 2014;69:234-40. [PMID: 23887867]
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nous dose of ceftriaxone or ertapenem, or a long-acting aminoglycoside is recommended before outpatient oral therapy. In a study comparing ciprofloxacin versus TMP-SMX, an initial dose of intravenous ceftriaxone resulted in improved outcomes in women receiving TMP-SMX who had a resistant uropathogen. Oral  $\beta$ -lactam agents are not recommended because of inferior efficacy rates.

In patients admitted for intravenous therapy, a broad-spectrum agent should be given until susceptibilities are known. A carbapenem agent, an extended-spectrum  $\beta$ -lactam with or without an aminoglycoside, or another regimen chosen on the basis of local resistance patterns is appropriate. Patients with risk factors for *Pseudomonas* or multidrug resistance who are severely ill should receive initial therapy with a carbapenem (imipenem-cilastatin, ertapenem, meropenem, or doripenem). This can be tailored once susceptibility is known. Two recently approved agents (ceftazidime-avibactam and ceftolozane-tazobactam) should be considered in cases of anticipated carbapenem resistance or extremely resistant *Pseudomonas*, typically in consultation with an infectious disease specialist.

Patients with underlying comorbidities and substantial prior health care exposure are more likely to have a drug-resistant organism, respond poorly to antimicrobial therapy even when the organism is susceptible, and develop complications if initial therapy is suboptimal. Broader-spectrum empirical therapy with agents to which resistance is least common and longer treatment durations are intended to blunt the negative effects of host compromise on treatment outcomes. In clinical trials of therapy for complicated UTI, oral fluoro-

quinolones were as effective as traditional intravenous regimens and at least as effective as oral TMP-SMX or TMP (particularly for organisms resistant to TMP-SMX). However, resistance to fluoroquinolones among uropathogens is increasing worldwide. If the local prevalence of fluoroquinolone resistance exceeds 10%, another broad-spectrum antimicrobial should be considered, including an extended-spectrum cephalosporin with or without an aminoglycoside or a carbapenem. Combinations of a  $\beta$ -lactam and a  $\beta$ -lactamase inhibitor (for example, ampicillin-sulbactam, ticarcillin-clavulanate, or piperacillin-tazobactam) or imipenem-cilastatin could also be considered (32). For CAUTI, the recommended duration of antimicrobial treatment is 7 days for patients who have prompt resolution of symptoms and 10-14 days for those with a delayed response. For those with CAUTI who are not severely ill, a 5-day regimen of levofloxacin may be considered (32, 54). In patients with CAUTI who have had a catheter in place for 2 weeks or longer, the catheter should be removed or replaced (32).

In pregnant women with symptomatic UTI, urine culture and susceptibility testing should be performed. Empirical therapy with an oral antimicrobial agent that is safe for use in pregnant women should be given for 3-7 days, depending on which drug is used. Antibiotic therapy should be adjusted on the basis of culture results.

### Is there a role for nonpharmacologic therapies, including alternative therapies?

There is no defined role for non-drug therapy in managing acute cystitis. Clinicians should advise patients that use of ibuprofen and nondrug approaches, such

as increased fluid intake, consumption of cranberry juice, and acupuncture, have no known benefit. Patients should be counseled that relying on nondrug therapies to treat acute cystitis is essentially the same as treating the condition with a placebo.

### **When should patients be hospitalized?**

Complicated cystitis or pyelonephritis may require inpatient management. Factors that may warrant hospitalization include serious comorbid conditions and pregnancy. Patients with high fever, dehydration, high leukocyte count, or other evidence of sepsis may require inpatient supervision, as do those who are vomiting and thus unable to take oral therapy. Another consideration is whether the patient has an upper urinary tract condition that requires drainage or surgical intervention. On rare occasions, a patient may have a multidrug-resistant organism in the urine that is susceptible only to parenterally administered antimicrobial agents or is intolerant of agents that can be given orally. Treatment trials have shown efficacy of oral therapy with a fluoroquinolone for mild, acute pyelonephritis in women, and thus many women with pyelonephritis can be managed successfully at home (55). Clinical judgment is required to determine which patients are appropriate for outpatient management, and a period of observation in the emergency department may be warranted.

### **What are common reasons for failure of therapy?**

Underlying medical conditions that increase risk for delayed or incomplete response to therapy, relapse, or infection include pregnancy, diabetes mellitus, and immunosuppression. Advanced age, recent UTI, recent antibiotic use, previous infection with a resistant organism, pro-

longed duration of symptoms before presentation (>3 days), and recent travel to an area with a high prevalence of antibiotic-resistant uropathogens all increase the risk for resistant infection. Urologic complications that increase risk for treatment failure include urinary tract stones, voiding disorders, indwelling catheters, stents, urinary obstruction, duplicated collecting system or other anatomical abnormalities, or vesicoureteral reflux. The combination of obstruction and infection puts the patient at increased risk for upper UTI (pyelonephritis), sepsis, and perinephric or intrarenal abscess. In particular, patients with CAUTI who do not respond promptly to appropriate antibiotic therapy may have obstruction or stones in the upper urinary tract. Effective urine drainage reduces intraluminal pressure and restores the flow of antibiotic-containing urine.

### **When should clinicians consult a specialist?**

Consultation with an infectious disease specialist is recommended for complicated UTI, particularly in patients with indwelling bladder catheters, in those hospitalized because of upper tract disease, in men with UTI, or when the organisms isolated in the urine are resistant to standard antibiotics.

Consultation with an infectious disease specialist or a urologist is warranted for patients with possible upper urinary tract involvement who do not respond to appropriate antibiotic therapy within 72 hours. A surgically correctable lesion may be present in men who report voiding difficulties or acute urine retention or who have early recurrent UTI or persistent microscopic hematuria. Stones, strictures, and occult cancer are among the differential diagnoses for men with these symptoms.

### **How should patients be followed?**

In uncomplicated cystitis, no follow-up is needed if symptoms resolve. In pregnant women, urine culture should be done after treatment to confirm eradication of bacteriuria, and repeated urinalyses or urine cultures should be done at intervals to confirm sterility of the urine through the time of delivery. Pregnant women who have already had bacteriuria during their pregnancy are at increased risk for recurrence and need closer monitoring than those who had sterile urine at initial screening. However, more evidence is needed to define the appropriate frequency of follow-up cultures and retreatment strategies (10).

Because "complicated UTI" is highly heterogeneous, the approach to monitoring during and after therapy must be individualized. Patients with complicated UTI should be followed clinically for resolution of symptoms and be reevaluated if symptoms worsen or do not improve within 24–48 hours. In CAUTI, it is important to monitor response to therapy by the patient's symptoms rather than by repeated urine cultures, given that recurrent bacteriuria is the norm in patients whose indwelling catheter remains in place.

### **What is the correct approach to secondary prevention in patients with a history of UTI?**

Recurrent UTI is estimated to occur in up to 50% of women within 1 year of an initial UTI (4). Other than the antimicrobial prophylaxis recommended in the Screening and Prevention section, there are few measures that women can take to prevent recurrent UTI. A large case-control study in premenopausal women found no associations between risk for recurrent UTI and precoital and postcoital voiding,

frequency of urination, wiping patterns, douching, use of hot tubs,

frequent use of pantyhose or tights, or daily fluid consumption (3).

**Treatment and Management...** The IDSA has standard-of-care guidelines for treatment of acute uncomplicated cystitis, acute uncomplicated pyelonephritis, and CAUTI. Fluoroquinolones should not be the first choice for uncomplicated cystitis given high levels of resistance and risk for serious adverse effects. Nonpharmacologic therapies for cystitis have no proven benefits and can lead to adverse outcomes. Posttreatment follow-up should include monitoring of the clinical response to therapy. Repeated urine cultures are needed only for pregnant women.

## CLINICAL BOTTOM LINE

## Practice Improvement

### What measures do U.S. stakeholders use to evaluate the quality of care?

Antibiotic stewardship has an essential role in slowing the rise of multidrug-resistant bacterial pathogens. In 2011, 262 million prescriptions were given for outpatient antibiotics in the United States (56). Data from cross-sectional outpatient surveys suggest that 30% of outpatient antibiotic prescriptions may be inappropriate (57). The principle of antibiotic stewardship espoused by the Centers for Disease Control and Prevention in its "Get Smart About Antibiotics" campaign that is relevant to treatment of UTI is to prescribe the right drug in the right dose for the right duration. Despite this, fluoroquinolones are the most commonly used antibiotic for uncomplicated acute cystitis (58–61). Prescribing antibiotics for an excessively long duration to treat acute cystitis is also a common practice (58, 62). Antibiotic stewardship programs for outpatient management of UTI must address both the type and duration of antimicrobial therapy. The urgency of addressing these issues was reinforced by an FDA warning in July 2016 (63) that fluoro-

quinolones should be avoided in uncomplicated UTIs (provided that alternative treatment options are available) because the risk for serious adverse effects generally outweighs the benefits.

The Joint Commission has added measuring and monitoring of CAUTI prevention practices and outcomes to its requirements for acute care hospitals (64), and consumers can compare CAUTI rates across hospitals by using Hospital Compare at Medicare.gov (65). The increased focus on prevention of CAUTI, particularly through decreasing unnecessary use of urinary catheters, has likely benefitted patients (66). For example, the Agency for Healthcare Research and Quality funded national campaigns in both acute and long-term care facilities that led to decreased use of urinary catheters and fewer CAUTIs (67, 68).

### What do professional organizations recommend with regard to care?

Professional organizations in the United States and abroad have released practice guidelines on UTI that address 3 categories: acute cystitis, CAUTI, and ASB. Guidelines for management and

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64. Requirements for the Catheter-Associated Urinary Tract Infections (CAUTI) National Patient Safety Goal for Hospitals. Accessed at [www.jointcommission.org/assets/1/6/R3\\_Cauti\\_HAP.pdf](http://www.jointcommission.org/assets/1/6/R3_Cauti_HAP.pdf) on 30 August 2017.
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prevention of CAUTI have been released by the Healthcare Infection Control Practices Advisory Committee of the Centers for Disease Control and Prevention (69), the Association for Professionals in Infection Control and Epidemiology in its Compendium of Strategies to Prevent Healthcare-Associated Infections in Acute Care Hospitals (70), the IDSA (32), and the European Association of Urology (71). ASB guidelines were published by the IDSA

in 2005, were endorsed by the U.S. Preventive Services Task Force in 2008, and are currently being updated. The Choosing Wisely campaign (72) of the American Board of Internal Medicine discourages urine culture in patients without symptoms localized to the urinary tract (that is, screening for ASB), recommends against treating bacteriuria in adults without specific urinary symptoms, and encourages removing or not placing unnecessary urinary catheters.

70. Yokoe DS, Anderson DJ, Berenholtz SM, Calfee DP, Dubberke ER, Ellingson KD, et al; Society for Healthcare Epidemiology of America (SHEA). A compendium of strategies to prevent healthcare-associated infections in acute care hospitals: 2014 updates. *Infect Control Hosp Epidemiol.* 2014;35:967-77. [PMID: 25026611]
71. European Association of Urology 2015 Guidelines on Urologic Infections. Accessed at [https://uroweb.org/wp-content/uploads/19-Urological-infections\\_LR2.pdf](https://uroweb.org/wp-content/uploads/19-Urological-infections_LR2.pdf) on 30 August 2017.
72. ABIM Foundation. AMDA - Urine cultures without clear signs and symptoms. 4 September 2013. Accessed at [www.choosingwisely.org/clinician-lists/amda-urine-cultures](http://www.choosingwisely.org/clinician-lists/amda-urine-cultures) on 17 August 2017.

## In the Clinic Tool Kit

### Urinary Tract Infection

#### Clinical Guidelines

[www.idsociety.org/Guidelines/Patient\\_Care/IDSA\\_Practice\\_Guidelines/Infections\\_by\\_Organ\\_System/Genitourinary/Uncomplicated\\_Cystitis\\_and\\_Pyelonephritis\\_\(UTI\)](http://www.idsociety.org/Guidelines/Patient_Care/IDSA_Practice_Guidelines/Infections_by_Organ_System/Genitourinary/Uncomplicated_Cystitis_and_Pyelonephritis_(UTI))

Guidelines from the Infectious Diseases Society of America.

[www.nice.org.uk/guidance/qs90](http://www.nice.org.uk/guidance/qs90)

Guidelines for treatment of adults with UTI from the National Institute for Health and Care Excellence.

#### Patient Information

[www.acponline.org/system/files/documents/practice-resources/patient-resources/urinary-tract-infection.pdf](http://www.acponline.org/system/files/documents/practice-resources/patient-resources/urinary-tract-infection.pdf)

Printable information for patients from the American College of Physicians.

<https://medlineplus.gov/urinarytractinfections.html>

Printable handouts for patients on various aspects of UTI.

In the Clinic

# WHAT YOU SHOULD KNOW ABOUT URINARY TRACT INFECTION

In the Clinic  
Annals of Internal Medicine

## What Is a Urinary Tract Infection?

A urinary tract infection (UTI) is a common infection of the urethra (the tube that takes urine out of the body), bladder, or kidney. Bacteria can get into the urethra and travel to the bladder, causing an infection. The infection can also travel to the kidneys. You may have higher risk for UTIs if you:

- Men can also get UTIs, but they are more common in women.
- Are sexually active.
- Are pregnant.
- Use certain types of birth control, such as a spermicide.
- Have a blocked flow of urine, for example from a kidney stone.
- Use a urinary catheter.
- Have diabetes.
- Have bladder or spinal cord injuries.
- Have an enlarged prostate.
- Have had a previous UTI.

## What Are the Warning Signs?

- Pain or burning when urinating
- Urgent or frequent need to urinate
- Blood in the urine
- Pain in the lower back or pelvic area
- Fever

## How Is It Diagnosed?

Your doctor will ask about your symptoms. In some cases, you may need to give a urine sample, which will help your doctor confirm UTI and decide which treatment is best.

## How Is It Treated?

- Medicines called antibiotics are used to treat UTIs.
- Some people get frequent UTIs. Talk with your doctor about treatment options if this is the case.

## How Can It Be Prevented?

If you have frequent UTIs, your doctor may suggest patient-initiated therapy when you have typical symptoms or may prescribe an antibiotic to take after sex, although it is unclear whether



this is a good option because resistance to antibiotics is increasing.

## Questions for My Doctor

- How can I prevent future UTIs?
- When will my symptoms go away?
- Do I need medicine? How should I take it?
- If I don't take antibiotics, will the infection go away on its own?
- What should I do if I don't feel better after taking the antibiotics?
- Does drinking cranberry juice help?
- Do over-the-counter medicines treat the infection?
- Should I stop having sex when I have a UTI?

## For More Information



### MedlinePlus

<https://medlineplus.gov/urinarytractinfections.html>

### National Institutes of Health

[www.niddk.nih.gov/health-information/urologic-diseases/bladder-infection-uti-in-adults](http://www.niddk.nih.gov/health-information/urologic-diseases/bladder-infection-uti-in-adults)